

CM0081 Formal Languages and Automata

§ 1.5 The Central Concepts of Automata Theory

Andrés Sicard-Ramírez

Universidad EAFIT

Semestre 2024-1

Preliminaries

Textbook

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman (2007). Introduction to Automata Theory, Languages, and Computation.

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Alphabets and Strings

Definition

An **alphabet** is a **finite**, non-empty set of symbols.

Alphabets and Strings

Definition

An **alphabet** is a **finite**, non-empty set of symbols.

Examples

$$\Sigma_1 = \{0, 1\},$$

$$\Sigma_2 = \{a, b, \dots, z\},$$

$$\Sigma_3 = \{x \mid x \text{ is a Unicode codepoint} \}.$$

Alphabets and Strings

Definition

A **string** (or **word** or **event**) is a **finite** sequence of symbols of an alphabet.

Alphabets and Strings

Definition

A **string** (or **word** or **event**) is a **finite** sequence of symbols of an alphabet.

Definition

The **empty string**, denoted by ε , is the string with **zero** occurrences of symbols.

Remark

The empty string may be chosen from **any** alphabet.

Alphabets and Strings

Definition

A **string** (or **word** or **event**) is a **finite** sequence of symbols of an alphabet.

Definition

The **empty string**, denoted by ε , is the string with **zero** occurrences of symbols.

Remark

The empty string may be chosen from **any** alphabet.

Conventions

Alphabets: $\Sigma, \Gamma, \dots$

Symbols: $a, b, c, \dots$

Strings: $w, x, y, z, \dots$

All Strings over an Alphabet

Definition

Let Σ be an alphabet. The **set of all the strings over Σ** (including the empty string), denoted Σ^* , can be inductively defined by the following clauses:

- (i) Basis step: $\varepsilon \in \Sigma^*$,
- (ii) Inductive step: If $x \in \Sigma^*$ and $a \in \Sigma$ then $xa \in \Sigma^*$.

Or, equivalently, by using the following inference rules:

$$\frac{}{\varepsilon \in \Sigma^*} \qquad \frac{x \in \Sigma^* \quad a \in \Sigma}{xa \in \Sigma^*}$$

Operations on Alphabets

Definition

Let a be a symbol on an alphabet Σ . The **powers of a** , denoted a^n , with $n \geq 0$, is the string formed by n repetitions of the symbol a (see, e.g. (Kozen 2012)). This operation is recursively defined by:

$$(-)^{(-)} : \Sigma \times \mathbb{N} \rightarrow \Sigma^*$$

$$a^0 = \varepsilon,$$

$$a^{n+1} = a^n a.$$

Operations on Strings

Definition

Let Σ be an alphabet. The **length** of a string x on Σ , denoted $|x|$ is the number of symbols in x . This function is **recursively** defined by

$$|-| : \Sigma^* \rightarrow \mathbb{N}$$

$$|\varepsilon| = 0,$$

$$|xa| = |x| + 1.$$

Operations on Strings

Definition

Let Σ be an alphabet. The **concatenation** of strings is **recursively** defined by

$$(-) \cdot (-) : \Sigma^* \times \Sigma^* \rightarrow \Sigma^*$$

$$x \cdot \varepsilon = x,$$

$$x \cdot ya = (x \cdot y)a.$$

That is, let $x = a_1a_2 \dots a_n$ and $y = b_1b_2 \dots b_n$ two strings, then

$$x \cdot y = a_1a_2 \dots a_m b_1b_2 \dots b_n.$$

Operations on Strings

Definition

Let Σ be an alphabet. The **concatenation** of strings is **recursively** defined by

$$(-) \cdot (-) : \Sigma^* \times \Sigma^* \rightarrow \Sigma^*$$

$$x \cdot \varepsilon = x,$$

$$x \cdot ya = (x \cdot y)a.$$

That is, let $x = a_1a_2 \dots a_n$ and $y = b_1b_2 \dots b_n$ two strings, then

$$x \cdot y = a_1a_2 \dots a_m b_1b_2 \dots b_n.$$

Notation

We remove the dot in the concatenation, that is, $xy := x \cdot y$.

Operations on Strings

Some properties of concatenation

Let x , y and z be strings.

- (i) Concatenation is associative, that is, $x(yz) = (xy)z$.
- (ii) The empty string is the unit for concatenation, that is, $x\varepsilon = \varepsilon x = x$.
- (iii) Concatenation is not commutative, that is, $xy \neq yx$.

Operations on Strings

Example

Let Σ be an alphabet and let x and y be strings over Σ . Prove that

$$|xy| = |x| + |y|.$$

Operations on Strings

Proof

By structural induction on y .

- Basis step ($y = \varepsilon$):

$$\begin{aligned} |x\varepsilon| &= |x| && \text{(def. of concatenation)} \\ &= |x| + |\varepsilon| && \text{(def. of length)} \end{aligned}$$

- Induction step ($y = wa$):

$$\begin{aligned} |x(wa)| &= |(xw)a| && \text{(def. of concatenation)} \\ &= |xw| + 1 && \text{(def. of length)} \\ &= (|x| + |w|) + 1 && \text{(IH)} \\ &= |x| + (|w| + 1) && \text{(arithmetic)} \\ &= |x| + |wa| && \text{(def. of length)} \end{aligned}$$



Operations on Strings

Proof

By structural induction on y (or by mathematical induction on $|y|$).

- Basis step ($y = \varepsilon$) (or $|y| = 0$, then $y = \varepsilon$):

$$\begin{aligned} |x\varepsilon| &= |x| && \text{(def. of concatenation)} \\ &= |x| + |\varepsilon| && \text{(def. of length)} \end{aligned}$$

- Induction step ($y = wa$) (or $|y| = n + 1$, then $y = wa$ where $|w| = n$):

$$\begin{aligned} |x(wa)| &= |(xw)a| && \text{(def. of concatenation)} \\ &= |xw| + 1 && \text{(def. of length)} \\ &= (|x| + |w|) + 1 && \text{(IH)} \\ &= |x| + (|w| + 1) && \text{(arithmetic)} \\ &= |x| + |wa| && \text{(def. of length)} \end{aligned}$$



Operations on Strings

Strings, length and concatenation in Haskell

```
data List    a = Nil | Cons a (List a)
data RList   a = Lin | Snoc (RList a) a
```

Operations on Strings

Strings, length and concatenation in Haskell

```
data List    a = Nil | Cons a (List a)
data RList   a = Lin | Snoc (RList a) a
```

```
lengthR :: RList a -> Int
lengthR Lin          = 0                -- Eq. 1
lengthR (Snoc xs x) = lengthR xs + 1    -- Eq. 2
```

Operations on Strings

Strings, length and concatenation in Haskell

```
data List    a = Nil | Cons a (List a)
data RList   a = Lin | Snoc (RList a) a
```

```
lengthR :: RList a -> Int
lengthR Lin          = 0                -- Eq. 1
lengthR (Snoc xs x) = lengthR xs + 1    -- Eq. 2
```

```
(+++ ) :: RList a -> RList a -> RList a
(+++) xs Lin          = xs              -- Eq. 1
(+++) xs (Snoc ys y) = Snoc (xs +++ ys) y -- Eq. 2
```

Operations on Strings

Example

Prove that $\text{lengthR } (xs \mathrel{+++} ys) = \text{lengthR } xs + \text{lengthR } ys$.

Operations on Strings

Example

Prove that $\text{lengthR } (xs \text{ +++ } ys) = \text{lengthR } xs + \text{lengthR } ys$.

Proof by structural recursion on ys

- Basis step (ys is Lin):

$$\begin{aligned} \text{lengthR } (xs \text{ +++ } \text{Lin}) & \\ &= \text{lengthR } xs && \text{(Eq. 1 of (+++))} \\ &= \text{lengthR } xs \text{ +++ } \text{lengthR } \text{Lin} && \text{(Eq. 1 of lengthR)} \end{aligned}$$

Operations on Strings

Proof by structural recursion on $|ys|$ (continuation)

- Induction step (ys is $\text{Snoc } ys' \ y'$):

$$\begin{aligned} & \text{lengthR } (xs \text{ +++ } (\text{Snoc } ys' \ y')) \\ &= \text{lengthR } (\text{Snoc } (xs \text{ +++ } ys') \ y') && \text{(Eq. 2 of (+++))} \\ &= \text{lengthR } (xs \text{ +++ } ys') + 1 && \text{(Eq. 2 of lengthR)} \\ &= (\text{lengthR } xs + \text{lengthR } ys') + 1 && \text{(IH)} \\ &= \text{lengthR } xs + (\text{lengthR } ys' + 1) && \text{(arithmetic)} \\ &= \text{lengthR } xs + \text{lengthR } (\text{Snoc } ys' \ y) && \text{(Eq. 2 of lengthR)} \end{aligned}$$



Operations on Strings

Definition

Let x be a string on an alphabet Σ . The **powers of x** , denoted x^n , with $n \geq 0$, is recursively defined by

$$(-)^{(-)} : \Sigma^* \times \mathbb{N} \rightarrow \Sigma^*$$

$$x^0 = \varepsilon,$$

$$x^{n+1} = x^n \cdot x.$$

Operations Alphabets

Definition

The **n-power** of an alphabet Σ , denoted Σ^n , is the set of strings of length n over Σ .

Examples

Given $\Sigma = \{0, 1\}$ then

$$\Sigma^0 = \{\varepsilon\},$$

$$\Sigma^1 = \{0, 1\},$$

$$\Sigma^2 = \{00, 01, 10, 11\},$$

$$\Sigma^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}.$$

Operations Alphabets

Example

Let Σ be an alphabet. Then

$$\Sigma^* = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$$

Languages

Definition

If Σ is an alphabet and $L \subseteq \Sigma^*$ then L is a **language** over Σ .

Languages

Definition

If Σ is an alphabet and $L \subseteq \Sigma^*$ then L is a **language** over Σ .

Examples

- $\emptyset$ and Σ^* are languages over any alphabet

Languages

Definition

If Σ is an alphabet and $L \subseteq \Sigma^*$ then L is a **language** over Σ .

Examples

- $\emptyset$ and Σ^* are languages over any alphabet
- The set of string of 0's and 1's with equal number of each

$$\{\varepsilon, 01, 10, 0011, 0110, 1001, 1100, \dots\}$$

Languages

Definition

If Σ is an alphabet and $L \subseteq \Sigma^*$ then L is a **language** over Σ .

Examples

- $\emptyset$ and Σ^* are languages over any alphabet
- The set of string of 0's and 1's with equal number of each

$$\{\varepsilon, 01, 10, 0011, 0110, 1001, 1100, \dots\}$$

- $\{0^n 1^n \mid n \geq 1\}$

Languages

Definition

If Σ is an alphabet and $L \subseteq \Sigma^*$ then L is a **language** over Σ .

Examples

- $\emptyset$ and Σ^* are languages over any alphabet
- The set of string of 0's and 1's with equal number of each

$$\{\varepsilon, 01, 10, 0011, 0110, 1001, 1100, \dots\}$$

- $\{0^n 1^n \mid n \geq 1\}$
- $\{\varepsilon\} \neq \emptyset$

Languages

Definition

If Σ is an alphabet and $L \subseteq \Sigma^*$ then L is a **language** over Σ .

Examples

- $\emptyset$ and Σ^* are languages over any alphabet
- The set of string of 0's and 1's with equal number of each

$$\{\varepsilon, 01, 10, 0011, 0110, 1001, 1100, \dots\}$$

- $\{0^n 1^n \mid n \geq 1\}$
- $\{\varepsilon\} \neq \emptyset$
- The set of binary numbers whose value is a prime

Languages

Definition

If Σ is an alphabet and $L \subseteq \Sigma^*$ then L is a **language** over Σ .

Examples

- $\emptyset$ and Σ^* are languages over any alphabet
- The set of string of 0's and 1's with equal number of each

$$\{\varepsilon, 01, 10, 0011, 0110, 1001, 1100, \dots\}$$

- $\{0^n 1^n \mid n \geq 1\}$
- $\{\varepsilon\} \neq \emptyset$
- The set of binary numbers whose value is a prime
- The set of legal C programs

Languages

Question

Is the set of legal English words a language?

Decision Problems

Definition

Let a set A the domain of a problem. A **decision problem** on A is a function (see, e.g. (Kozen 2012))

$$f : A \rightarrow \{0, 1\}.$$

Decision Problems

Definition

Let $L \subseteq \Sigma^*$ be a language and let $w \in \Sigma^*$ be string. The **decision problem for L** is to decide whether or not $w \in L$.

Decision Problems

Definition

Let $L \subseteq \Sigma^*$ be a language and let $w \in \Sigma^*$ be string. The **decision problem for L** is to decide whether or not $w \in L$.

Some questions

- (i) Is it a problem or a decision problem?

Decision Problems

Definition

Let $L \subseteq \Sigma^*$ be a language and let $w \in \Sigma^*$ be string. The **decision problem for L** is to decide whether or not $w \in L$.

Some questions

- (i) Is it a problem or a decision problem?
- (ii) Is it a language or a problem?

Decision Problems

Definition

Let $L \subseteq \Sigma^*$ be a language and let $w \in \Sigma^*$ be string. The **decision problem for L** is to decide whether or not $w \in L$.

Some questions

- (i) Is it a problem or a decision problem?
- (ii) Is it a language or a problem?
- (iii) Is the problem decidable or undecidable?

Decision Problems

Definition

Let $L \subseteq \Sigma^*$ be a language and let $w \in \Sigma^*$ be string. The **decision problem for L** is to decide whether or not $w \in L$.

Some questions

- (i) Is it a problem or a decision problem?
- (ii) Is it a language or a problem?
- (iii) Is the problem decidable or undecidable?
- (iv) Is the problem tractable or intractable?

References

- John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).
- Dexter C. Kozen [1997] (2012). Automata and Computability. Third printing. Undergraduate Texts in Computer Science. Springer. DOI: [10.1007/978-1-4612-1844-9](https://doi.org/10.1007/978-1-4612-1844-9) (cit. on pp. 9, 34).